

# **The Levelized Cost of Energy**

**Comparing how much a kilowatt-hour *really* costs**

PHY199-RAP · Energy in the 21st Century

# What is LCOE?

The **levelized cost of energy** is the average price per unit of electricity that pays back *all* lifetime costs of a plant.

$$\text{LCOE} = \frac{\text{lifetime costs (build + fuel + upkeep)}}{\text{lifetime energy produced}}$$

It lets us compare a solar farm, a gas plant, and a reactor on the **same footing** — dollars per megawatt-hour.

# A simple calculation

A solar farm costs **\$1,000,000** to build and runs for **25 years**, producing **2,500 MWh per year**.

$$\text{LCOE} = \frac{\$1,000,000}{25 \times 2,500 \text{ MWh}} = \$16 / \text{MWh}$$

*(Real estimates add fuel, maintenance, and a discount rate — but the idea is the same.)*

# Rough LCOE by source

| Source        | Typical LCOE (\$/MWh) |
|---------------|-----------------------|
| Utility solar | 30 – 40               |
| Onshore wind  | 25 – 50               |
| Natural gas   | 45 – 75               |
| Nuclear       | 55 – 90               |
| Coal          | 65 – 150              |

Cheapest on paper isn't the whole story — **land, waste, and human cost** don't appear here.

# Takeaway

LCOE is a powerful *first* comparison —  
but a fair energy choice weighs the costs it **leaves out**.